



III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	C	D	B	C	C	A	D	C	C	C	A	B	D	B

16	17	18	19	20	21	22	23	24	25	26				
C	C	D	D	D	C	D	B	D	A	A				

- 1 A Electric potential difference

$$\Delta V = \frac{W}{Q} = \frac{P}{I}$$

- 2 C Fraction of power delivered to lamp

$$\frac{P_L}{P_T} = \frac{I^2 R}{I^2 R + I^2 r} = \frac{R}{R + r}$$

- 3 D Since the electrons and protons are moving in opposite directions, the total current is the sum of the current due to both charges.

$$I = \left(\frac{dn_e}{dt} + \frac{dn_p}{dt} \right) \times e = (7 \times 10^{18} + 2 \times 10^{18}) \times 1.60 \times 10^{-19} = 1.44 \text{ A}$$

- 4 B Using $I = nAve$ and $L = v\Delta t$

$$L = \frac{I\Delta t}{nAe} = \frac{1.0 \times 60}{8.5 \times 10^{28} \times 3.2 \times 10^{-7} \times 1.60 \times 10^{-19}} = 0.014 \text{ m}$$

- 5 C Charge $Q = \frac{1}{2}(I_i + I_f)\Delta t = \frac{1}{2}(0.100 + 0.020) \times 8.0 = 480 \text{ mC}$

- 6 C Current in rod

$$I = nAvq = 8.5 \times 10^{28} \times 8.0 \times 10^{-6} \times 2.3 \times 10^{-5} \times 1.60 \times 10^{-19} = 2.5 \text{ A}$$

- 7 A Power in circuit

$$P = I^2 R = \frac{E^2 R}{(R + 3r)^2} \propto E^2$$

So

$$\frac{\text{power in R in circuit (i)}}{\text{power in R in circuit (ii)}} = \left(\frac{E_{(i)}}{E_{(ii)}} \right)^2 = \left(\frac{4.5}{1.5} \right)^2 = 9$$

- 8 D Drift velocity

$$v = \frac{I}{nAq} = \frac{0.15}{7.8 \times 10^{28} \times \pi \left(\frac{1}{2} \times 0.35 \times 10^{-3} \right)^2 \times 1.60 \times 10^{-19}} = 1.2 \times 10^{-4} \text{ m s}^{-1}$$

- 9 C Root-mean-square current

$$I_{\text{rms}} = \sqrt{\frac{4.0^2 \times 1.0 + 2.0^2 \times 2.0}{4.0}} = 2.45 \text{ A}$$

Power dissipated

$$P = I_{\text{rms}}^2 R = 2.45^2 \times 11 = 66 \text{ W}$$

- 10 C Root-mean-square voltage

$$V_{\text{rms}} = \sqrt{\frac{4.0^2 \times 1.0 + 1.0^2 \times 3.0}{5}} = 1.95 \text{ V}$$

Power dissipated

$$P = \frac{V_{\text{rms}}^2}{R} = \frac{1.95^2}{2.0} = 1.9 \text{ W}$$

- 11 C A fuse breaks when its temperature exceeds a pre-determined value.

The temperature of the wire depends on the mean power dissipated. So a fuse breaks when the current (whether d.c. or a.c. current) exceeds the rated value.

In this case, the rated current is 13 A d.c. or 13 A r.m.s.

- 12 A Since $\langle P \rangle \propto V_{\text{rms}}^2$

$$V_{\text{rms},2} = \sqrt{\frac{\langle P \rangle_2}{\langle P \rangle_1}} \times V_{\text{rms},1} = \sqrt{\frac{3P}{P}} \times 4.0 = 6.9 \text{ V}$$

since r.m.s voltage is the constant voltage that produces the same average power dissipation in the same resistor.

- 13 B Mean power in case 1

$$P = \frac{1}{2} I_0^2 R \quad \square$$

Mean power in case 2

$$4P = I_{\text{rms}}^2 \times 2R \quad \square$$

$\square \div \square :$

$$4 = 4 \times \frac{I_{\text{rms}}^2}{I_0^2} \quad \text{p } I_{\text{rms}} = I_0$$

- 14 D The p.d. across AB is half-wave modified.

$$V_{0,AB} = 2V_{\text{rms},AB} = 2 \times 2.5V_{\text{rms,NB}} = 12.5 \text{ V}$$

Peak power dissipated across AB

$$P_{0,AB} = \frac{V_{0,AB}^2}{R} = \frac{12.5^2}{10} = 16 \text{ W}$$

- 15 B Since $P \propto V^2$,

$$P_2 = \left(\frac{V_2}{V_1}\right)^2 \times P_1 = \left(\frac{120}{240}\right)^2 \times 2.0 = 0.50 \text{ kW}$$

Energy consumed in 1.5 hours

$$E_2 = P_2 t = 0.50 \times 1.5 = 0.75 \text{ kWh}$$

- 16 C Mean power in Fig A

$$W = \frac{V_0^2}{2R}$$

Mean power in Fig. B

$$P = \frac{\frac{1}{2} \times 4V_0^2}{R} = \frac{2V_0^2}{R} = 4W$$

- 17 C Since $\langle P \rangle = I_{\text{rms}} V_{\text{rms}}$, r.m.s. current would range from

$$\frac{\langle P \rangle_{\min}}{V_{\text{rms,max}}} \leq I_{\text{rms}} \leq \frac{\langle P \rangle_{\max}}{V_{\text{rms,min}}}$$

$$8.33 \text{ A} \leq I_{\text{rms}} \leq 10.9 \text{ A}$$

$$11.8 \text{ A} \leq I_0 \leq 15.4 \text{ A}$$

Hence probably current, $I = 14.1 \sin(375t)$.

Note: The angular frequency ω is irrelevant to the calculation.

From $\omega = 2\pi f$, angular frequency would range from 314 rad s^{-1} to 377 rad s^{-1} .

- 18 D Since both d.c. and a.c. have the same heating effect,

$$\frac{1}{2} I_0^2 \times \frac{1}{4} R = I^2 R \quad \text{p } I_0 = 2\sqrt{2}I$$

- 19 D For sinusoidal a.c., mean power

$$\langle P \rangle = \frac{1}{2} P_0 = \frac{V_0^2}{2R}$$

- 20 D Peak voltage

$$V_0 = 2 \times 5.00 = 10.0 \text{ V}$$

Period

$$T = 4 \times 1.0 = 4.0 \text{ s}$$

So equation of waveform

$$V = V_0 \sin \omega t = 10.0 \sin \left(\frac{2\pi}{4.0} \cdot t \right) = 10 \sin(1.6t)$$

- 21 C For both d.c. and a.c. have the same heating effect, $I_{\text{dc}} = I_{\text{rms}}$.

Peak value of alternating current

$$I_0 = \sqrt{2} I_{\text{rms}} = \sqrt{2} I_{\text{dc}} = \sqrt{2} \times 3.0 = 4.2 \text{ A}$$

- 22 D For a half-wave rectified sinusoidal a.c., peak current $I_0 = 2I$.

- 23 B Since $\langle P \rangle \propto V_{\text{rms}}^2$, power dissipated per bulb

$$\langle P_2 \rangle = \frac{V_{\text{rms},2}^2}{V_{\text{rms},1}^2} \times \langle P_1 \rangle = \frac{V_{\text{rms},2}^2}{\frac{1}{2} V_{10}^2} \times \langle P_1 \rangle = \frac{120^2}{\frac{1}{2} \times 150^2} \times 13 = 16.6 \text{ W}$$

So total power dissipated

$$\langle P_{2T} \rangle = 2 \langle P_2 \rangle = 33 \text{ W}$$

Since root-mean-square voltage across each bulb is 120 V,

$$V_{20} = \sqrt{2} V_{\text{rms}2} = \sqrt{2} \times 120 = 170 \text{ V}$$

- 24 D The p.d. across the load is half-wave rectified.

$$V_0 = 2V_{\text{rms}} = 2 \times 80 = 160 \text{ V}$$

Peak power dissipated across load

$$P_0 = \frac{V_0^2}{R} = \frac{160^2}{10} = 2560 \text{ W}$$

Note: Power dissipated is independent of frequency of a.c.

- 25 A Since $\langle P \rangle \propto V_{\text{rms}}^2$

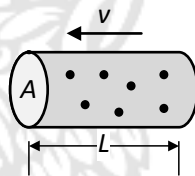
$$\langle P_2 \rangle = \left(\frac{V_{\text{rms},2}}{V_{\text{rms},1}} \right)^2 \times \langle P_1 \rangle = \left(\frac{120}{240} \right)^2 \times 600 = 150 \text{ W}$$

- 26 A Both the sinusoidal and square wave (with the same peak voltage) has the same root-mean-square voltage. Hence both voltages have the same mean rate of heat produced.

Structured Questions

1. [ACJC/H2/Prelims 2018/P2/5]

a. i.



Current

$$I = \frac{dQ}{dt} = e \frac{dN}{dt} = e \frac{d(nAL)}{dt} = nAve$$

where L is the length of the section of electrons that passes through some cross-sectional area of the wire per unit time.

ii. Since both wires are connected in series, the same current flows through the wires.

Since drift velocity $v \propto \frac{1}{nA}$,

$$v_W = \frac{n_{Cu} A_{Cu}}{n_W A_W} \times v_{Cu} = \frac{n_{Cu} d_{Cu}^2}{n_W d_W^2} \times v_{Cu} = \frac{8.0 \times 10^{28} \times (1.4 \times 10^{-3})^2}{3.4 \times 10^{28} \times (0.020 \times 10^{-3})^2} \times 0.021 \times 10^{-3} = 0.24 \text{ m s}^{-1}$$

2. [PJC/H2/Prelims 2018/P3/5]

a. The root-mean-square voltage is that value of a steady direct voltage that would produce thermal energy at the same rate in a given resistor.

*b. i. Turns ratio

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{240 \times 10^3}{12.0 \times 10^3} = 20$$

ii. Transmission current

$$I = \frac{P}{V} = \frac{20.0 \times 10^6}{240 \times 10^3} = 83.3 \text{ A}$$

Power loss during transmission

$$P_{\text{loss}} = I^2 R = 83.3^2 \times 200 = 1.39 \times 10^6 \text{ W}$$

iii. Energy loss in one day

$$E_{\text{loss}} = 1.39 \times 10^6 \times 24 = 3.34 \times 10^7 \text{ Wh} = 33400 \text{ kWh}$$

Amount of money loss = $33400 \times \$0.10 = \3340

3. [YJC/H2/Prelims 2018/P3/7]

a. i. The root-mean-square voltage is that value of a steady direct voltage that would produce thermal energy at the same rate in a given resistor.

ii. Peak power dissipated in lamp

$$P_0 = 2\langle P \rangle = 2I_{\text{rms}} V_{\text{rms}} = 2 \times 230 \times 0.26 = 120 \text{ W}$$

b. Root-mean-square voltage

$$V_{\text{rms}} = \sqrt{\frac{10.0^2 \times 2.0 + 5.0^2 \times 1.0}{4.0}} = 7.5 \text{ V}$$